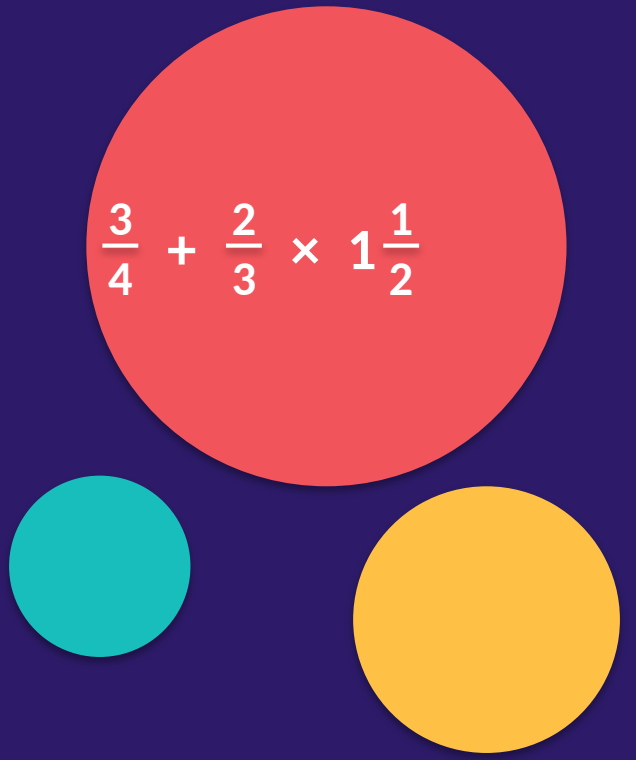
Year 8 • OnMaths tutoring

Fractions in Context: Worded Problems



Add • Subtract • Multiply • Divide • Area • Mean

Lesson for Isabella • 60 minutes

Have ready: mini whiteboard, pen, calculator OFF

Today's objectives

1. Solve worded problems that need adding or subtracting fractions
2. Solve worded problems that need multiplying or dividing fractions
3. Find areas of shapes with fraction lengths
4. Find the mean of a set of fractions

Starter: warm up the skills

Do now • 5 min

a) $\frac{2}{3} + \frac{1}{4}$

$= \frac{11}{12}$

b) $\frac{5}{6} - \frac{1}{3}$

$= \frac{3}{6} = \frac{1}{2}$

c) $\frac{3}{5} \times \frac{10}{9}$

$= \frac{30}{45} = \frac{2}{3}$

d) $\frac{4}{7} \div \frac{2}{3}$

$= \frac{4}{7} \times \frac{3}{2} = \frac{6}{7}$

Decode the words

Key idea • 4 min

Key words → operation

altogether, total, sum, more than **+**

left, remain, difference, how much more **−**

of, each, per, times, area **×**

share, how many fit / can be cut, per each **÷**

4 steps every time
1. Read – underline the numbers
2. Draw – bar model or sketch
3. Decide – which operation?
4. Answer – simplify + units

Which operation?

A pizza is cut up. Ben eats $\frac{3}{8}$ and Ali eats $\frac{1}{4}$. How much is eaten?

+ (total eaten)

How many $\frac{1}{4}$ kg bags can be filled from 3 kg?

÷ (how many fit)

Find $\frac{2}{5}$ of 35 m.

× ("of" means times)

Objective 1: Add & subtract — I do

I do • 4 min

Ella spends $\frac{1}{4}$ of her pocket money on snacks and $\frac{2}{5}$ on a book.

What fraction of her money does she have left?

Bar model (20 equal parts)



snacks book left

Working

Spent: $\frac{1}{4} + \frac{2}{5}$

$= \frac{5}{20} + \frac{8}{20}$

$= \frac{13}{20}$

Left: $1 - \frac{13}{20} = \frac{20}{20} - \frac{13}{20}$

She has $\frac{7}{20}$ left

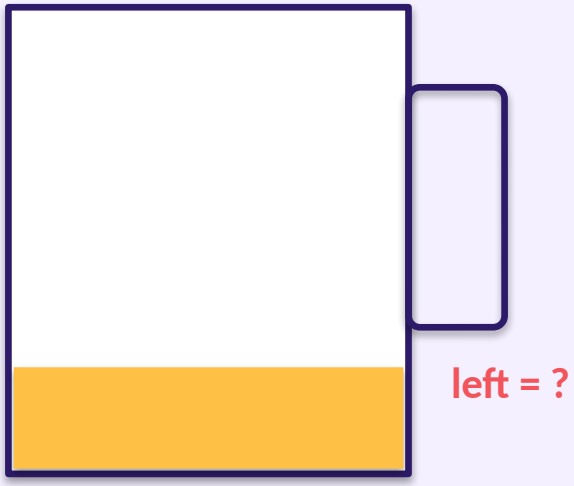
Objective 1: Add & subtract — We do

We do • 4 min

A jug holds $2\frac{1}{2}$ litres of juice.

Tom pours out $1\frac{3}{4}$ litres.

How much juice is left in the jug?



Working

$2\frac{1}{2} - 1\frac{3}{4}$

$= \frac{5}{2} - \frac{7}{4}$

$= \frac{10}{4} - \frac{7}{4}$

$= \frac{3}{4}$ litre left

Objective 1: Mini whiteboard

You do • 4 min

1 Amir walks $1\frac{2}{3}$ km to school, then $\frac{3}{4}$ km to the shop.

How far does he walk altogether?

$\frac{29}{12} = 2\frac{5}{12}$ km

2 A ribbon is $3\frac{1}{2}$ m long. $1\frac{5}{6}$ m is cut off.

How much ribbon remains?

$\frac{10}{6} = 1\frac{2}{3}$ m

3 In a class, $\frac{3}{8}$ walk to school and $\frac{1}{3}$ come by bus. The rest come by car. What fraction come by car?

$1 - \frac{17}{24} = \frac{7}{24}$

4 A water tank is $\frac{5}{6}$ full. After a shower it is $\frac{3}{8}$ full. What fraction of the tank was used?

$\frac{20}{24} - \frac{9}{24} = \frac{11}{24}$

Objective 2: Multiply & divide — I do

I do • 5 min

A bag of flour weighs $2\frac{1}{4}$ kg.

How much do $\frac{2}{3}$ of the bag weigh?

Working

$\frac{2}{3} \times 2\frac{1}{4}$

$= \frac{2}{3} \times \frac{9}{4}$

$= \frac{18}{12} = \frac{3}{2}$

$= 1\frac{1}{2}$ kg

A plank is $4\frac{1}{2}$ m long.

How many $\frac{3}{4}$ m shelves can be cut from it?



Working

$4\frac{1}{2} \div \frac{3}{4}$

$= \frac{9}{2} \times \frac{4}{3}$

$= \frac{36}{6}$

= 6 shelves

Objective 2: Mini whiteboard

You do • 5 min

1 A snail crawls $\frac{3}{8}$ m every minute.

How far does it crawl in $2\frac{2}{3}$ minutes?

$\frac{3}{8} \times \frac{8}{3} = 1$ m

2 $5\frac{1}{4}$ pizzas are shared equally between 7 friends. How much pizza does each get?

$\frac{21}{4} \div 7 = \frac{3}{4}$ pizza

3 How many $\frac{2}{3}$ litre bottles can be filled from 10 litres of water?

$10 \div \frac{2}{3} = 15$ bottles

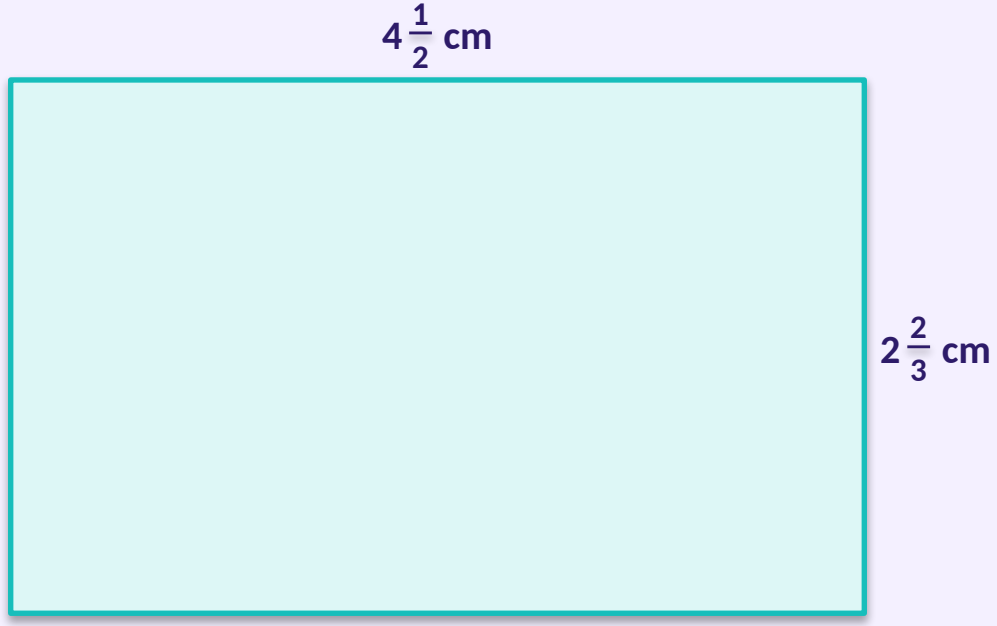
4 A cycle route is $12\frac{1}{2}$ km. Mia has ridden $\frac{3}{5}$ of it. How far has she ridden?

$\frac{3}{5} \times \frac{25}{2} = 7\frac{1}{2}$ km

Objective 3: Area — I do

I do • 3 min

Find the area of this rectangle.



Working

Area = length × width

$= 4\frac{1}{2} \times 2\frac{2}{3}$

$= \frac{9}{2} \times \frac{8}{3}$

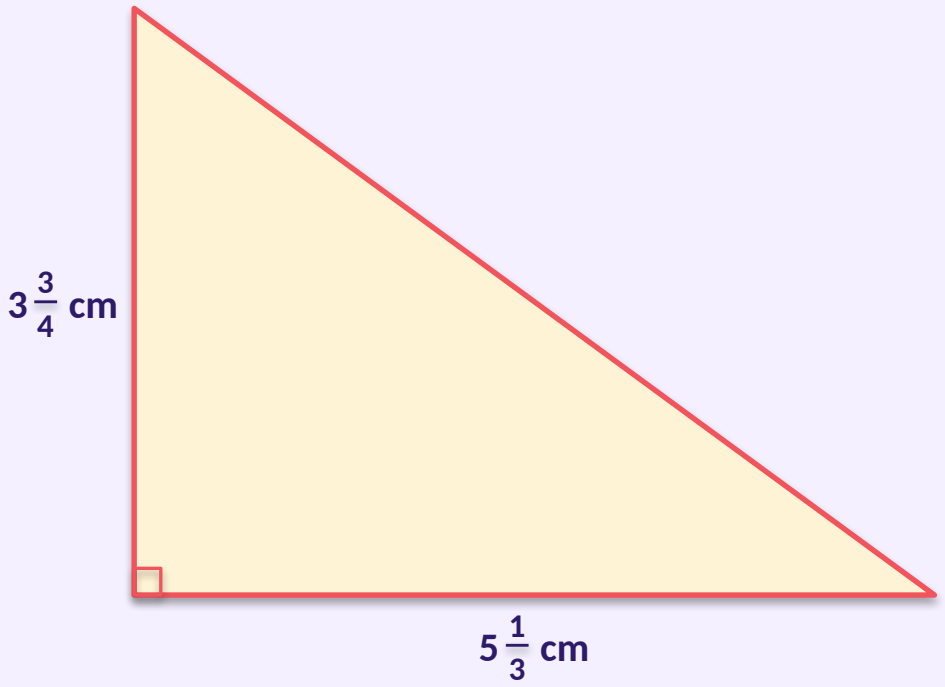
$= \frac{72}{6}$

$= 12\ \text{cm}^2$

Objective 3: Area of a triangle — We do

We do • 3 min

Find the area of this right-angled triangle.



Working

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times \frac{16}{3} \times \frac{15}{4}$$

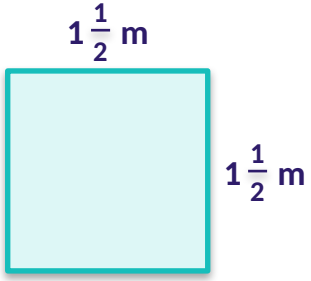
$$= \frac{240}{24}$$

$$= 10 \text{ cm}^2$$

Objective 3: Mini whiteboard

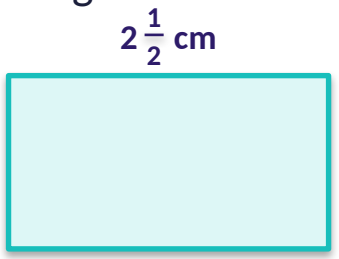
You do • 5 min

1 Find the area of the square.



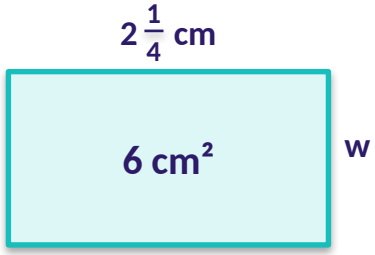
$\frac{3}{2} \times \frac{3}{2} = \frac{9}{4} = 2\frac{1}{4}$ m²

2 Find the area of the rectangle.



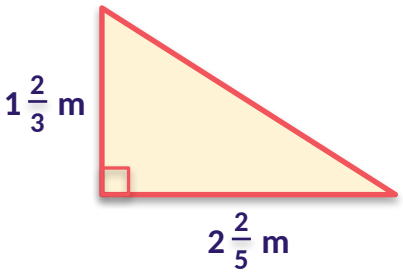
$\frac{5}{2} \times \frac{8}{5} = 4$ cm²

3 The area is 6 cm². Find the width w.



$6 \div \frac{9}{4} = \frac{8}{3} = 2\frac{2}{3}$ cm

4 Find the area of the triangle.



$\frac{1}{2} \times \frac{12}{5} \times \frac{5}{3} = 2$ m²

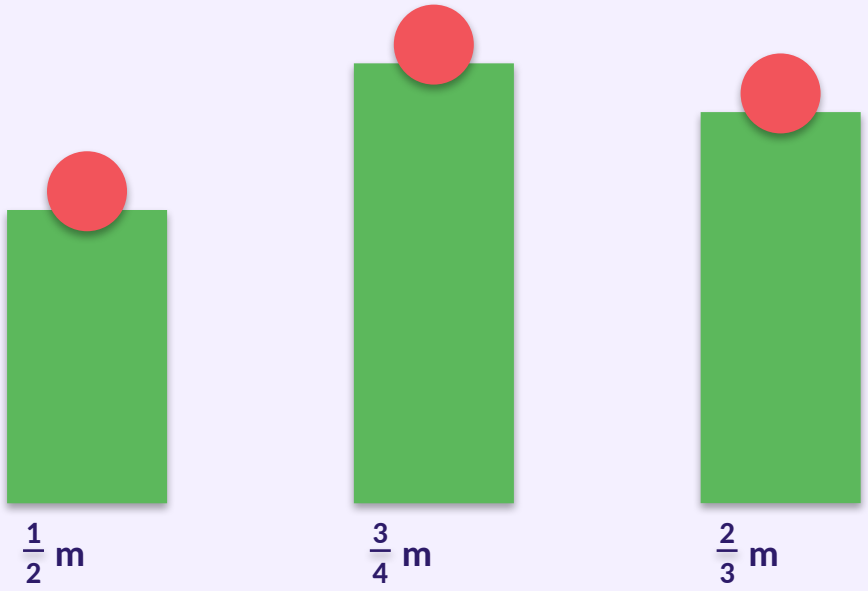
Objective 4: Mean — I do

I do • 3 min

Three tomato plants are $\frac{1}{2}$ m, $\frac{3}{4}$ m and $\frac{2}{3}$ m tall.

Find the mean height.

Mean = total ÷ how many



Working

Total: $\frac{1}{2} + \frac{3}{4} + \frac{2}{3}$

$= \frac{6}{12} + \frac{9}{12} + \frac{8}{12} = \frac{23}{12}$

Mean: $\frac{23}{12} \div 3 = \frac{23}{12} \times \frac{1}{3}$

Mean = $\frac{23}{36}$ m

Objective 4: Mean — We do

We do • 3 min

Kai runs on four days. Find the mean distance he runs per day.

Mon	$2\frac{1}{2}$ km
Tue	$1\frac{3}{4}$ km
Wed	3 km
Thu	$2\frac{1}{4}$ km

Working

Total: $2\frac{1}{2} + 1\frac{3}{4} + 3 + 2\frac{1}{4}$

$= 9\frac{1}{2}$ km

Mean: $\frac{19}{2} \div 4 = \frac{19}{2} \times \frac{1}{4}$

$= \frac{19}{8} = 2\frac{3}{8}$ km

Objective 4: Mini whiteboard

You do • 4 min

1 Find the mean of $\frac{1}{3}$ and $\frac{1}{2}$.

$\frac{5}{6} \div 2 = \frac{5}{12}$

2 Find the mean of $\frac{3}{4}$, $\frac{5}{8}$ and $\frac{1}{2}$.

$\frac{15}{8} \div 3 = \frac{5}{8}$

3 Four bags weigh $1\frac{1}{2}$ kg, $2\frac{1}{4}$ kg, $1\frac{3}{4}$ kg and $2\frac{1}{2}$ kg. Find the mean weight.

$8 \div 4 = 2$ kg

4 The mean of three numbers is $\frac{3}{4}$. Two of them are $\frac{1}{2}$ and $\frac{5}{6}$. Find the third number.

$\frac{9}{4} - \frac{4}{3} = \frac{11}{12}$

Stretch: multi-step problems

OPTIONAL • if time

A garden is $7\frac{1}{2}$ m by 4 m. $\frac{2}{5}$ of it is lawn. The rest is split equally into 3 flower beds. Find the area of each flower bed.

Working

Area $= \frac{15}{2} \times 4 = 30$ m²

Lawn $= \frac{2}{5} \times 30 = 12$ m²

Rest $= 18$ m²

Each bed $= 18 \div 3 = 6$ m²

Maya spends $\frac{1}{3}$ of her money on Monday and $\frac{1}{4}$ of what is left on Tuesday. She has £24 left. How much did she start with?

Working

After Monday: $\frac{2}{3}$ left

Tuesday: $\frac{1}{4} \times \frac{2}{3} = \frac{1}{6}$

Left: $\frac{2}{3} - \frac{1}{6} = \frac{1}{2}$

$\frac{1}{2}$ = £24 so she started with £48

Exit ticket

Assess • 5 min

1 A bottle holds $1\frac{3}{4}$ litres of water. Jo drinks $\frac{2}{3}$ of it. How much does she drink?

$\frac{7}{4} \times \frac{2}{3} = \frac{7}{6} = 1\frac{1}{6}$ litres

2 A rectangle is $3\frac{1}{3}$ cm long and $2\frac{2}{5}$ cm wide. Find its area.

$\frac{10}{3} \times \frac{12}{5} = 8\ \text{cm}^2$

3 Find the mean of $1\frac{1}{2}$, $\frac{2}{3}$ and $\frac{5}{6}$.

$3 \div 3 = 1$

How did you do?

Review • 1 min

1	I can solve worded problems that need adding or subtracting fractions	✓	~	?
2	I can solve worded problems that need multiplying or dividing fractions	✓	~	?
3	I can find areas of shapes with fraction lengths	✓	~	?
4	I can find the mean of a set of fractions	✓	~	?

Remember: Read → Draw → Decide the operation → Answer with units